

CS2220: Introduction to Computational Biology

Basic Local Alignment Search Tool

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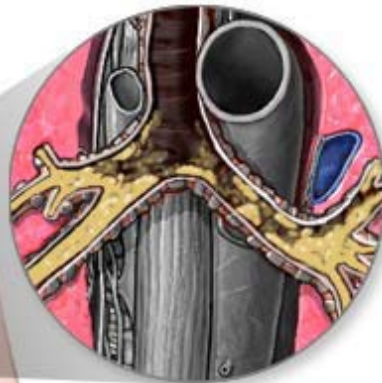
Materials for this presentation have been adapted from the following sources:

NCBI HelpDesk - Field Guide Course Materials

Bioinformatics: A practical guide to the analysis of genes and proteins

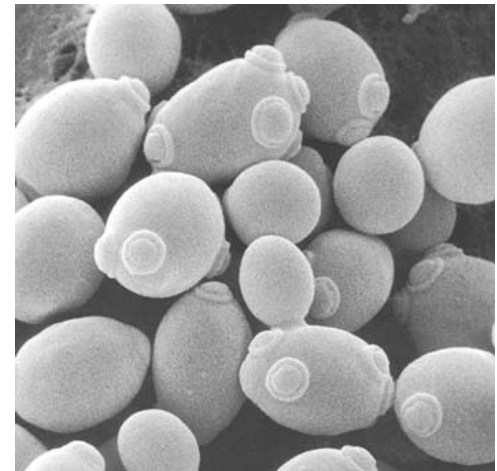
Finding Model Organisms for Study of Disease

Can **yeast** be used as a **model organism** to study **cystic fibrosis**?



Cystic fibrosis is a hereditary disorder characterized by lung congestion and infection and malabsorption of nutrients by the pancreas

ADAM.

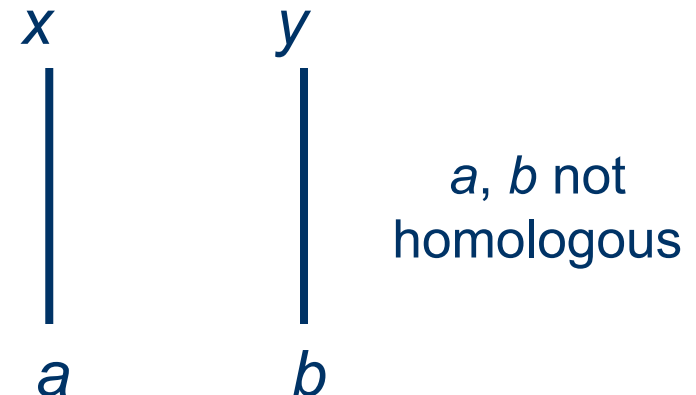
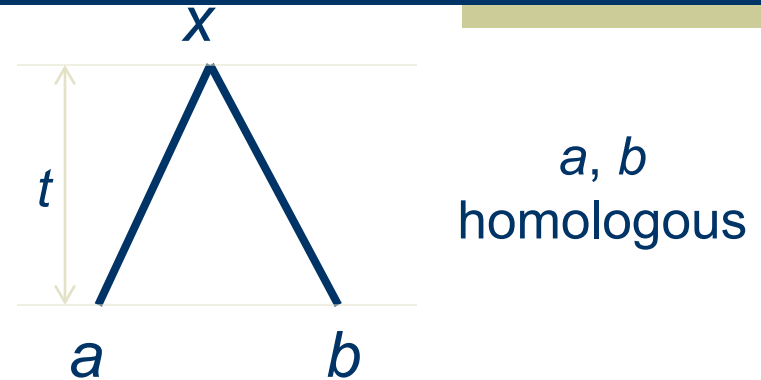


Model Organisms

- **Cystic fibrosis is a genetic disorder that affects humans**
 - If **yeast** contain a protein that is related (**homologous**) to the protein involved in cystic fibrosis
 - Then **yeast can be used as a model organism** to study this disease
 - **Study of the protein in yeast will tell us about the function of the protein in humans**

Sequence Homology

- Homologous protein or DNA sequences share common ancestry
 - A statement of homology is therefore an *evolutionary hypothesis*
- Homology need not imply similar function
- Homology is a binary property, a pair of sequences are either *homologous* or *not homologous*.
 - No such thing as *degree of homology*
- Homology is often inferred by sequence similarity

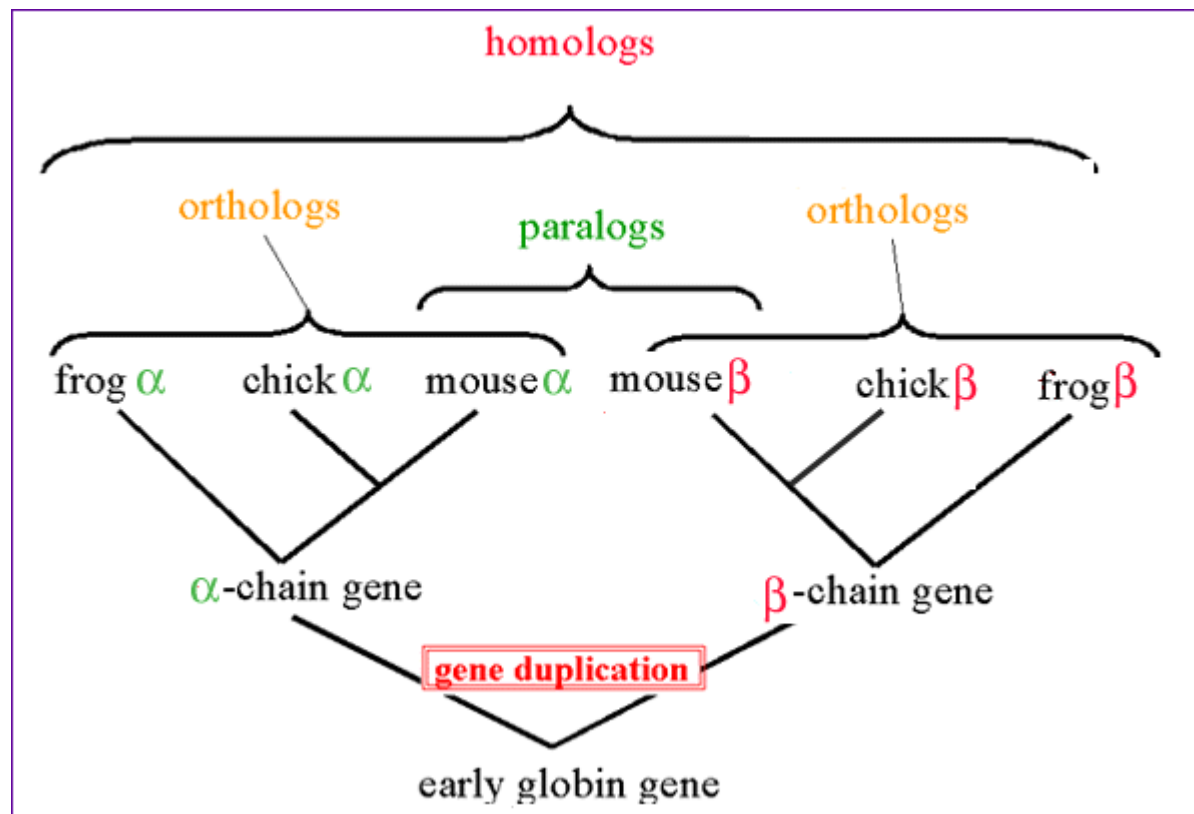


Orthology and paralogy

"Where the homology is a result of gene **duplication** so that both copies have descended side by side during the history of an organism, (for example, alpha and beta hemoglobin) the genes should be called **paralogous** (para=in parallel). Where the homology is the result of **speciation** so that the history of the gene reflects the history of the species (for example alpha hemoglobin in man and mouse) the genes should be called **orthologous** (ortho=exact). "

Fitch WM. *Distinguishing homologous from analogous proteins*. Systematic Zoology 1970 Jun;**19**(2):99-113.

Orthology, paralogy and multigene families



Reproduced from NCBI education website

GOAL: to find **homologous** genes
and proteins

Homologous Proteins (or genes)

- Have a common ancestor (they're related)
 - Have similar structures
 - Have similar functions

Criteria for considering two sequences to be homologous

- **Proteins are homologous if**
 - Their amino acid sequences are at least **25% identical**
- **DNA sequences are homologous if**
 - they are at least **70% identical**
 - Note that sequences must be over 100 a.a. (or bp) in length

**Whenever possible, it is better to
compare proteins
than to compare genes**

WHY????

Similarity Scores

- We want these scores to distinguish the related sequences from the unrelated sequences
- So we select alignment parameters for database searching that give us the **best distinguishing scores**
- These may not be the parameters that will give us the most accurate alignment

Similarity Scores

- Similarity scores - PAM, Blosum similarity matrices
 - Rescue us from having to assume that all amino acid changes are equally likely and equally harmful
 - Different similarity matrices are appropriate for different degrees of evolutionary divergence
 - More later

Computer Algorithm

- Dynamic programming is most sensitive and the least selective (WHAT DOES IT MEAN??)
- BLAST and FASTA are much faster and more selective which can be an advantage
- **No program is always best at finding distantly related sequences for all gene or protein families**, but dynamic programming is guaranteed to give optimal alignment for given scores

Database search assumptions

- The sequences sought have an evolutionary ancestral sequence in common with the “query sequence”
- All substitutions are not equally likely and should be weighted to account for this
- Insertions and deletions are **less likely** than substitutions and should be weighted to account for this.

Database search

Basic principles of Database searching

- Search query sequence against all sequence in database
- Calculate score and select top sequences
- Dynamic programming is best

Approximation Algorithms

- Heuristic approximation to Smith-Waterman
 - Runs faster
 - Loses some sensitivity

Database search FASTA

Approximation Algorithms

- ✓ Fast sequence search
- ✓ Based on dotplot
- ✓ Identify identical words (k-tuples)
- ✓ Search significant diagonals
- ✓ Use PAM 250 for further refinement
- ✓ Dynamic programming for narrow region

FASTA Algorithm

1. Divide query sequence into its constituent overlapping words of length (e.g.) two for proteins or (e.g.) six for nucleic acids
2. Each sequence in the database is also broken up in the same way
3. Two word lists are compared to find all identical words in both sequences
4. An initial score is computed based on how many identities are found within a small region of the dot plot.

FASTA algorithm, cont

- First, look for all identities between small “word” = ktup and every sequence in database. Ktup size determines how many letters must be identical (e.g., 3)

- CTGCACTA **AGCTGACGCA**
- CTG **CTG**
- TGC **GCA**
- GCA
- etc

FASTA, cont

- Ktup matches can be depicted in a matrix; diagonals indicate matches.
- The top 10 diagonals are rescored using substitution matrices and identities smaller than ktup; each of these rescored diagonals is called an initial region.

	a	g	c	t	g	a	c	g	c	a
C							-		-	
T										
G		-						-		
C			-				-		-	
A	-					-				-
C			-				-		-	
T			-							
A	-					-				-

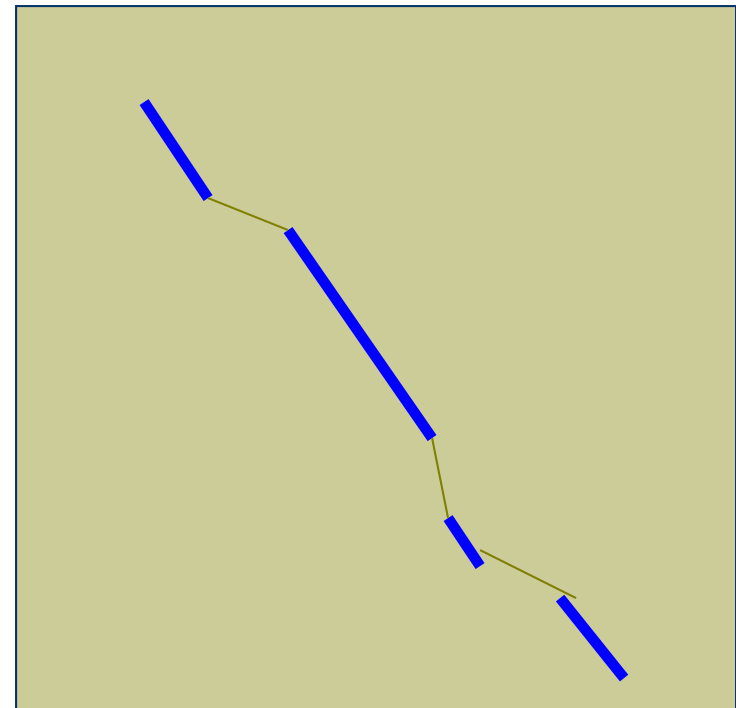
FASTA Algorithm

5. If the initial score is high enough, a second score is computed by evaluating which initial identities can be joined into a consistent alignment using only gaps of less than the window size.

WHAT IS THIS STEP TRYING TO ACHIEVE?

FASTA, cont.

- Initial regions are then joined using a joining penalty (like a gap penalty). The highest score of the joined initial regions is then the score for that library sequence. The library sequences are ranked by this score.



FASTA Algorithm

6. If the secondary score is high enough, then a Smith-Waterman alignment is performed within the same region of the dot plot defined by the concentrated identities and using the same window-size.
7. This third score is reported as the optimal score.

FASTA Summary

- Uses local alignment and then, extends it to the global alignment.
- Good for similarity searches in less similar sequences

WHY????

- Too sensitive
 - Reduced speed
- } as a result of alignment procedure

Database search BLAST

- Basic Local Alignment Search Tool

Approximation Algorithms

- ✓ Heuristic method to find the highest scoring
 - ✓ Locally optimal alignments
- Why BLAST is popular?
 - Good balance of sensitivity and speed
 - Reliable
 - Flexible
 - Produce local alignments: short significant stretches of similarity, irrespective of where they are in the sequence

BLAST compares sequences

- **BLAST** takes a **query** sequence
- **Compares** it with millions of sequences in the **Genbank databases**
 - By constructing local alignments
- **Lists** those that appear to be similar to the query sequence
 - The “**hit list**”
- **Tells you why** it thinks they are **homologous**
 - **BLAST** makes suggestions
 - **YOU** make the conclusions

BLAST Programs

The most common BLAST search include
five programs:

Program	Database (Subject)	Query
BLAST N	Nucleotide	Nucleotide
BLAST P	Protein	Protein
BLAST X	Protein	Nt. ➔ Protein
T BLAST N	Nt. ➔ Protein	Protein
T BLAST X	Nt. ➔ Protein	Nt. ➔ Protein

BLASTN

- **BLASTN**
 - The query is a nucleotide sequence
 - The database is a nucleotide database
 - No conversion is done on the query or database
- **DNA :: DNA homology**
 - Mapping oligos to a genome
 - Annotating genomic DNA with transcriptome data from ESTs and RNA-Seq
 - Annotating untranslated regions

BLASTP

- **BLASTP**
 - The query is an amino acid sequence
 - The database is an amino acid database
 - No conversion is done on the query or database
- **Protein :: Protein homology**
 - Protein function exploration
 - Extract novel gene

BLASTX

- **BLASTX**
 - The query is a nucleotide sequence
 - The database is an amino acid database
 - All six reading frames are translated on the query and used to search the database
- **Coding nucleotide seq :: Protein homology**
 - Gene finding in genomic DNA
 - Annotating ESTs and transcripts assembled from RNA-Seq data

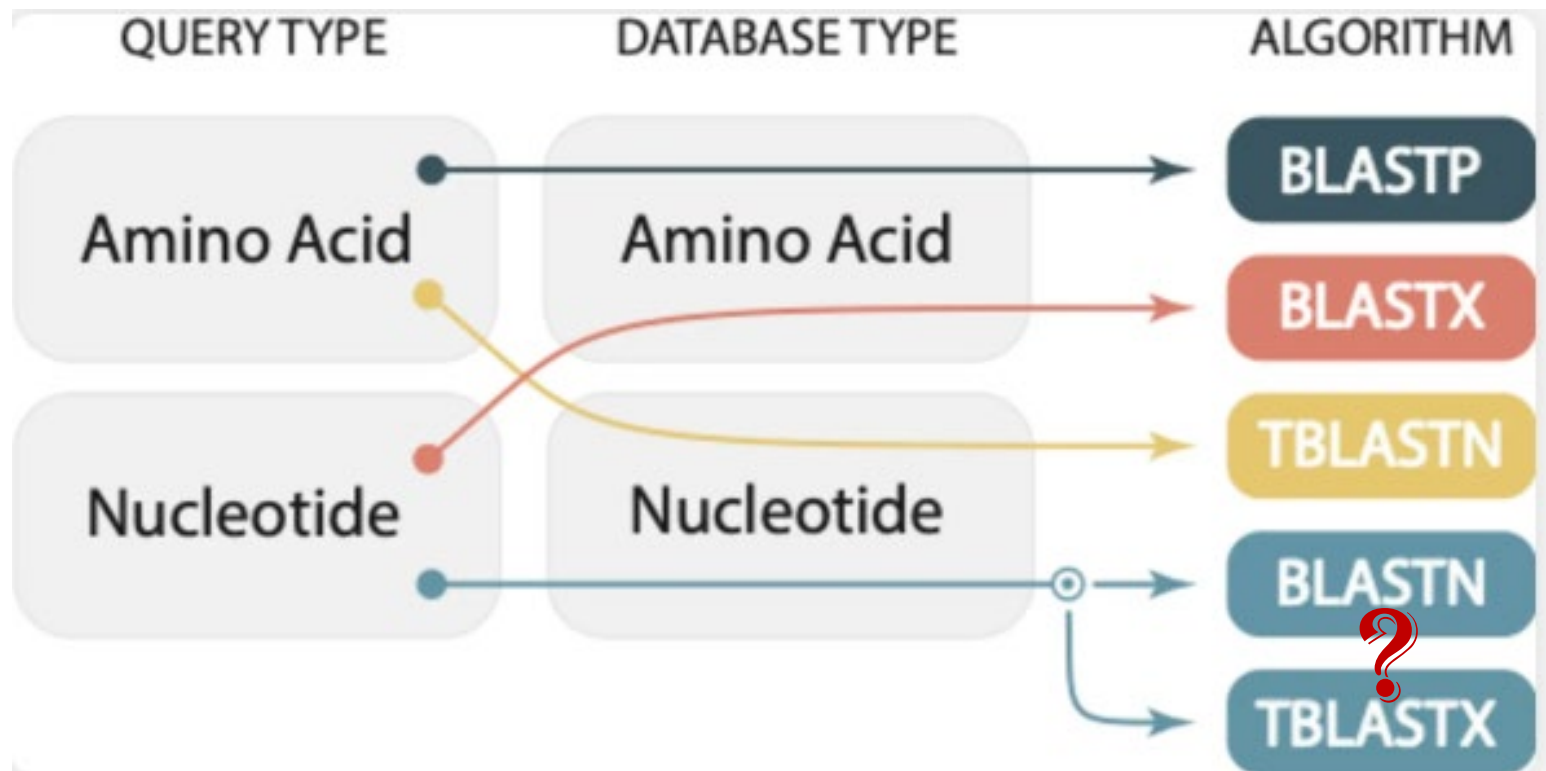
TBLASTN

- **TBLASTN**
 - The query is an amino sequence
 - The database is a nucleotide database
 - All six frames are translated in the database and searched with the protein sequence
- **Protein :: Coding nucleotide DB homology**
 - Mapping a protein to a genome
 - Mining ESTs and RNA-Seq data for protein similarities

TBLASTX

- TBLASTX
 - The query is a nucleotide sequence
 - The database is a nucleotide database
 - All six frames are translated on the query and on the database
- Coding :: Coding homology
 - Searching distantly-related species
 - Sensitive but expensive

BLAST



BLASTN vs. TBLASTX

- BLASTN is faster because it makes far fewer comparisons, and each comparison is more straightforward than TBLASTX.
- BLASTN is more precise for highly similar nucleotide sequences.
- TBLASTX is more sensitive for divergent sequences. It can better detect similarity among distantly related sequences.

$4 * 4 * 4 = 64$ POSSIBLE CODONS VS. 20 AMINO ACIDS

What does BLAST do?

BLAST Algorithm

- Three heuristic layers:
 1. Seeding – identify where to start alignment
 - Given a query q and a target sequence, find substrings of length k (k -mers) of score at least t --- also called *hits*. k is normally 3 to 5 for amino acids and 12 for nucleotides.
 2. Extension – extending alignment from seeds
 - Extend each hit to a *locally maximal segment*. Terminate the extension when the reduction in score exceeds a pre-defined threshold
 3. Evaluation – Determine which alignments are statistically significant
 - Report maximal segments above score S .

1. Seeding

- Idea: significant alignments have *words* in common
 - *Word* is a defined number of letters
 - Example: MGQLV contains 3-letter words MGQ, GQL, QLV
- BLAST locates all common words in a pair of sequences, then uses them as seeds for the alignment
- Eliminates a lot of the search space

1. Seeding, cont

- Make a list of short “words” = wmers from the query sequence (same as FASTA).
 - Choose all possible wmers by sliding a window of length w down the query sequence
- Then it expands this list in order to recover sensitivity lost by only using exact matches.

1. Seeding, Ex.

DNA
 CACTAGCTAAA
 For $w = 6$.
 CACTAG
 ACTAGC
 CTAGCT
 etc.

Protein
 WRKRKKRTGLE
 For $w=3, T=11$
 WRK
 WRR
 WKR
 RKR
 RKK, etc.

What is a word hit?

- Simple definition: two identical words
- In practice, some good alignments may not contain identical words
- *Neighborhood* – all words that have a high similarity score to the word – at least as big as a threshold T
- Higher values of T reduce number of hits
- Word size W also affects number of hits
- Adjusting T and W controls both speed and sensitivity of BLAST

Seeding, Ex.

Query Word ($W = 3$)

TLSHAWRLSNETDKRPFIEAERL**RDQ**HKKDYPEYKYQPRRRKNGKPGSSSEADAHSE

Determine neighborhood

RDQ 16	QDQ 12	EDQ 11	RDN 11	RDB 11	BDQ 10	RDP 10
RBQ 14	REQ 12	HDQ 11	RDD 11	ADQ 10	XDQ 10	RDT 10
RDZ 14	RDR 12	ZDQ 11	RDH 11	MDQ 10	RQQ 10	RDY 10
KDQ 13	RDK 12	RNQ 11	RDM 11	SDQ 10	RSQ 10	RDX 10
RDE 13	NDQ 11	RZQ 11	RDS 11	TDQ 10	RDA 10	DDQ 9 ...

All possible words, start with TLS, LSH, SHA, HAW, ...

Scoring of matches done using scoring matrices

All words above a threshold are deemed to be similar to the original query word
 $T=11$

query word RDQ would be similar to REQ because it's above threshold.

Where does the score (S) come from?

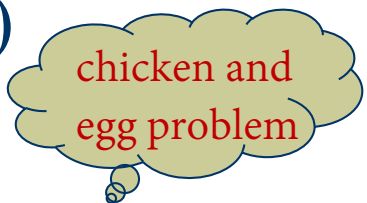
- Substitution matrices are used for amino acid alignments.
 - each possible residue substitution is given a score
- A simpler unitary matrix is used for DNA pairs (+1 for match, -2 mismatch)

	A	C	D	E	F	G	H	
A	4	0	-2	-1	-2	0	-2	
C	0	9	-3	-4	-2	-3	-3	
D	-2	-3	6	2	-3	-1	-1	
E	-1	-4	2	5	-3	-2	0	
F	-2	-2	-3	-3	6	-3	-1	
G	0	-3	-1	-2	-3	1	-2	
H	-2	-3	-1	0	-1	-2	0	

BLOSUM 62

Scoring Matrices

- PAM (**P**ercent **A**ccepted **M**utation)
 - Theoretical approach
 - Based on assumptions of mutation probabilities
- BLOSUM (**B**LOcks **S**Ubstitution **M**atrix)
 - Empirical
 - Constructed from multiply aligned protein families
 - Ungapped segments (blocks) clustered based on percent identity



chicken and
egg problem

Which Similarity Matrix to Use?

- In database searching, a PAM or Blosum matrix corresponding to an inappropriate degree of divergence can cause you to fail to discover homologous sequences that are present in the database.
- Therefore a thorough database search will involve using at least 2 and most likely 3 different matrices.
- Using different matrices usually has a higher payoff than using different programs and search algorithms.

2. Extension

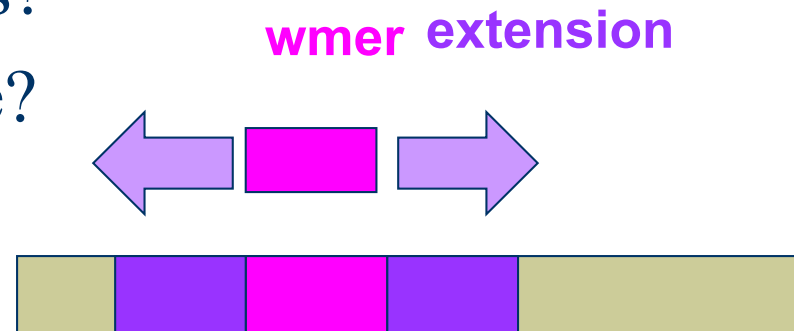
- Once search space is seeded, alignments generated by extending in both directions from the seed

- Example:

The quick brown fox jumps over the lazy dog.

The quiet brown cat purrs when she sees him.

- Can align first six characters?
- How far should we continue?



2. Extension, cont

- *X* parameter – How much is score allowed to drop off after last maximum?
 - Extend hits; any extensions must increase the score over that of the original hit; extension stops when no increase in score
- Example (assume identical scores +1 and mismatch scores -1)

The quick brown fox jump

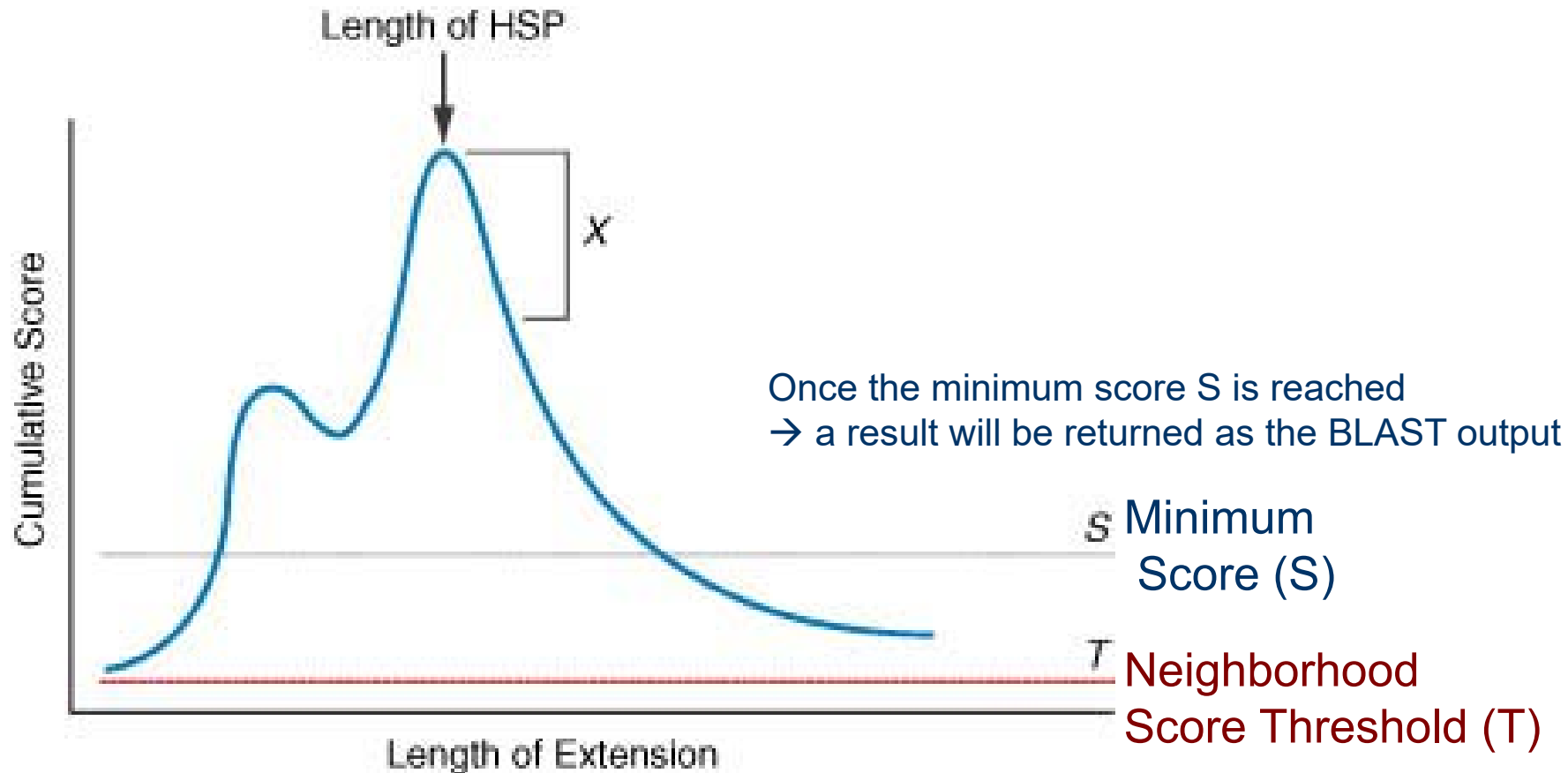
The quiet brown cat purr

123 45654 56789 876 5654 <-score

000 00012 10000 123 4345 <-drop off

Extending the High Scoring Segment Pair (HSP)

addition for matches, negative for mismatches



What is a good value of X ?

- Small X risks premature termination
- But little benefit to a very large X
 - So little information that they are unlikely to be critical in finding a maximal segment pair alignment.
- Generally better to use large value
- W and T better for controlling speed than X

3. Evaluation

- To find a locally optimal alignment : BLAST accumulates the score as the alignment is extended in both directions.
- When a run of mostly negative scores is encountered, the cumulative score will drop substantially → it is unlikely that the score will rebound.
 - This observation provides the basis for an additional heuristic whereby the extension of a hit is terminated when the reduction in score exceeds a dropoff threshold.
- The local alignment with the highest score is returned.

Example

BLAST Algorithm

1. Remove low complexity sections of the sequence

- These are areas of common or repeated sequence which will be matched and confuse BLAST

2. Make a k-letter word list of the sequence

- K is variable, but usually 11 for nucleotide sequences
- Sequentially apply a 'window' of length k over the sequence

q l n f s a g w	Sequence
q l	Word 1
l n	Word 2
n f	Word 3
f s	.
s a	.
a g	.
g w	

BLAST Algorithm , cont

3. Compare this word-list with the list of all k-length words and, using the substitution matrix, score words according to their substitution
4. Repeat for all words in the sequence – accept only those words over the threshold

Word Expanded List

ql: ql, qm, hl, zl

ln: ln, lb

nf: nf, af, ny, df, qf, ef, gf, hf, kf, sf, tf, bf, zf

fs: fs, fa, fn, fd, fg, fp, ft, fb, ys

sa: no words score 8 or more, including the initial word sa

ag: ag

gw: gw, aw, rw, nw, dw, qw, ew, hw, iw, kw, mw, pw, sw,

tw, vw,bw, zw, xw

BLAST Algorithm , cont

5. Compare the revised wordlist with the database sequences

Query sequence: R P P Q G L F

Database sequence: D P P E G V V

└─▶ Exact match is scanned.

Score: -2 7 7 2 6 1 -1

└─▶ HSP

Optimal accumulated score = $7+7+2+6+1 = 23$

6. Extend the matches to find the highest scoring pair (HSP)
7. Return the sequences that exceed some threshold

Brief Comparison of BLAST and FASTA

Attribute	BLAST	FASTA
Algorithm	uses a heuristic algorithm to find local sequence alignments.	uses a dynamic programming algorithm to find local sequence alignments.
Database	search against various databases, e.g. NCBI's GenBank, Swiss-Prot	search against custom databases created by the user.
Speed	generally faster than FASTA due to its heuristic approach.	slower compared to BLAST due to its dynamic programming approach.
Scoring	uses a scoring system based on sequence similarity and alignment quality.	uses a scoring system based on sequence similarity and alignment quality.
Output	provides a detailed report with alignment statistics, E-values, etc.	provides a simple text-based output with sequence alignments.

**How do I input a query
into BLAST?**

Enter your “query” sequence

- A sequence can be input as a (an)
 - **FASTA** format sequence
 - **Accession number**
- Protein blast can only accept amino acid sequences

Choose search set

- Choose which database to search
 - **Default** is non-redundant protein sequences (nr)
 - Searches all databases that contain protein sequences

Choose organism

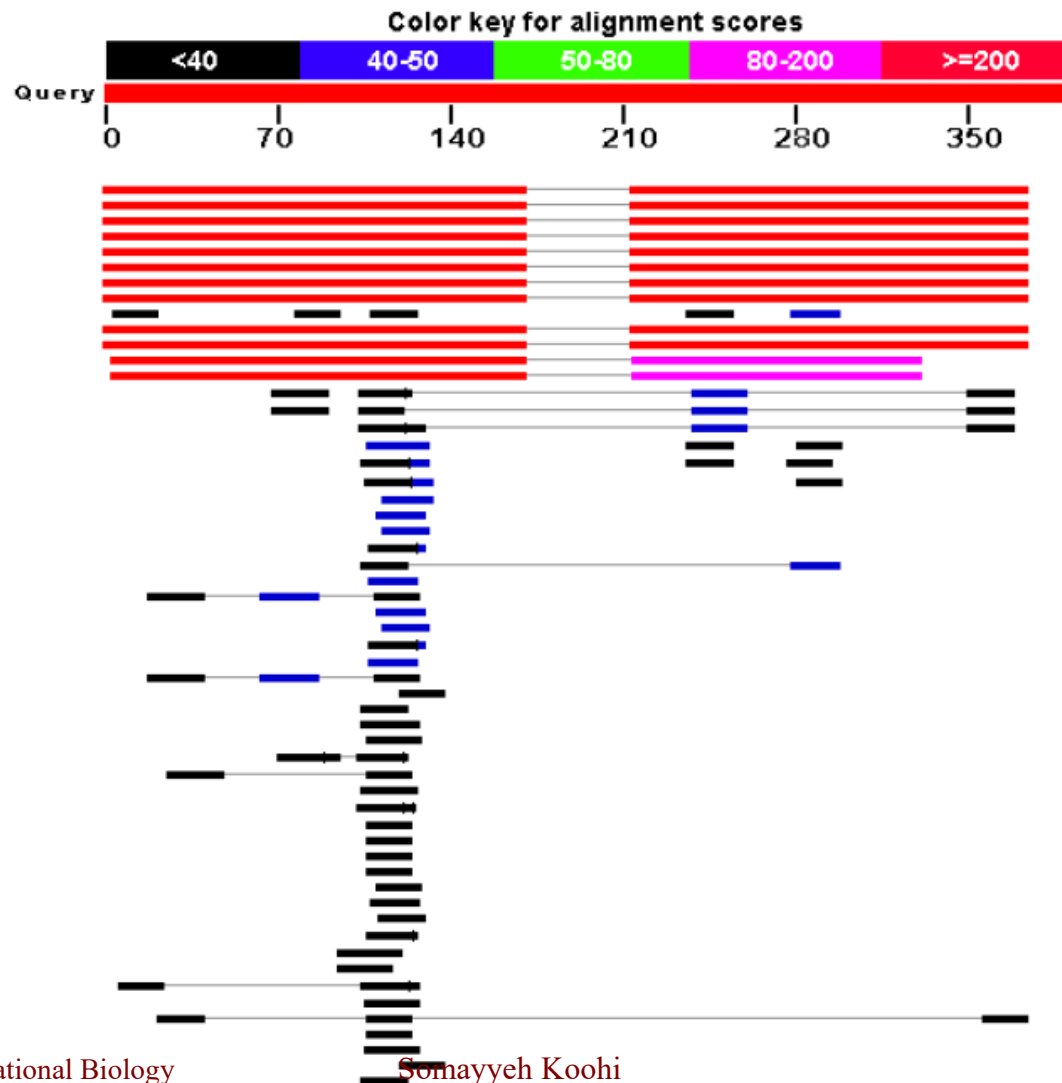
- **Default** is all organisms represented in databases
- Use this to limit your search to one organism (eg. Yeast)

BLAST off!!

- Click on the **BLAST** button at the bottom of the page!

How do I interpret the results of a BLAST search?

1. Graphical view



2. Hit list

- **BLAST lists the best matches (hits)**
 1. **Matching Sequences:** The sequences from the database that align with the query sequence.
 2. **E-values:** The expectation values (E-values) indicate the likelihood of the match occurring by chance.
 1. Lower E-values suggest more significant alignments.
 3. **Bit Scores:** A measure of the quality of the alignment.
 1. Higher bit scores indicate better alignments.
 4. **Percent Identity:** The percentage of identical matches between the query and subject sequences in the aligned region.
 5. **Alignment Information:** Details of how the query sequence aligns with the subject sequence, including mismatches, gaps, and alignment length.
 6. **Accession Numbers:** Identifiers for the database sequences that were aligned with the query.

**Most people use the E- value as their
first indication of similarity!**

Look at E-value
Look at graphic display
If necessary, look at alignment

Make your best guess!

References

- http://www.psc.edu/biomed/TUTORIALS/SEQUENCE/DB_SEARCH/tutorial.html
- <http://www.ncbi.nlm.nih.gov/BLAST/>
- Flexible Sequence Similarity Searching with the FASTA3 program package, William Pearson
- Bioinformatics A Practical Guide the Analysis of Genes and Proteins Baxevanis & Ouellette